Jason B. Hill

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Introduction

I transform organizations with data. I handle data from soup to nuts. I understand business needs, engineer a pipeline, analyze data, forecast outcomes, and (most importantly) drive change. I focus on predictability over short-term gains.

Technical Stack

Languages & Frameworks: Python (Jupyter, NumPy, Pandas, Dash, SciPy), Clojure, Bash, R, Java (Micronaut, Spring)

Data & Infrastructure: Google Cloud/GCP (BigQuery, Firestore, Functions, Logging, Pub/Sub, GCS, CX-Agents, Vertex AI, Composer, Looker/DataStudio), PostgreSQL (window functions, UDFs, PL/pgSQL, recursive CTEs), AWS, CI/CD, Jenkins, Apache Airflow, Kafka, Splunk, Spark, PySpark, Spark SQL, Docker, Oracle, architecture (Kimball/star, snowflake, xNF)

Forecasting: Monte Carlo, Bayesian inference, random forests, XGBoost

AI/ML: Vertex AI / GCP Conversational Agents, OpenAI, scikit-learn, TensorFlow, RAG, Cursor, Claude Code

Professional Experience

Rally Software — Data Engineer, Senior Data Engineer
CA Agile Central — Data Scientist, Senior Data Scientist, Principal Data Scientist
Broadcom (Agile Operations Division) — Staff Data Scientist
Broomfield, Denver, Boulder, CO | Oct 2015 – Oct 2025

- Directed forecasting products. Organized inter-team development, QA, and testing for product releases.
- Developed a Smart Cumulative Flow Diagram for analyzing team and train delivery metrics.
- Led train-level portfolio flow metrics design and development.
- Used Vertex AI (Gemini) to produce flow metrics chart interpretations and coaching advice for teams and trains.
- Spearheaded capacity and delivery forecasting development for team and train-level planning.
- Developed, maintained, and used an idempotent distributed tracing lambda architecture data pipeline for a visible and reliable usage and performance metrics system to report 100% of user interactions in a SaaS environment involving 250K+ monthly users. Used the resulting data with Splunk, BigQuery, Looker, Spark, and other systems to analyze interactions. Held two years of long-lambda data for future needs. Data averaged 400 GiB per day.
- Led performance optimization, via tracing and visualizations, for Rally's Oracle-to-PostgreSQL migration.
- Built and maintained a "product data hub" for those within Rally to track feature adoption, subscription usage, performance, and user input.
- Built a leading forecast indicator for subscription renewal predictions using usage data and random forest models.
- I left a legacy here. I transformed the organization using data.

SpotXChange — Data Scientist
Broomfield, CO | Apr 2014 – Sep 2015

Silicon Graphics, Inc. (SGI) — Software Engineer
Longmont, CO | Jun 2013 – Mar 2014

Education

Ph.D. (ABD) in Mathematics — University of Colorado (advisors: Alexander Hulpke and Keith Kearnes) | 2014

- Computational group and graph theory, algebraic combinatorics, supercomputing

M.A. in Mathematics — University of Colorado (advisor: Keith Kearnes) | 2010

- Computational group theory, representation theory

M.S. in Mathematics — University of Vermont (advisor: David S. Dummit) | 2006

- Computational algebraic number theory

B.S. in Mathematics — University of Michigan–Flint | 2004

- Mathematics

About Me

Other Work

- I've taught many mathematics courses at the university level. I've taught several graduate-level courses.
- Adjunct instructor: Saint Michael's College, Mathematics, 2006
- Adjunct instructor: Community College of Vermont, Photography, 2006
- Contract photographer: Automobile Magazine, location shoots, 2002–2005

Volunteering

- Over the winter holidays, I work with the Save Elephant Foundation and Elephant Nature Park in northern Thailand to rescue and rehabilitate abused elephants. My wife and I sponsor multiple elephants.
- A Precious Child in Broomfield, CO.
- Mount Washington Obeservatory.

Random Stuff

- I've climbed 28 of Colorado's 14ers. Overall, I've summited 14ers 39 times.
- Since moving to Colorado, I've played in more than 800 softball games. I've played every position except pitcher.
- I've been to 51 countries.
- I rode bike trials in the Vans Warped Tour three times.
- I'm face-blind. I don't recognize people, but my math brain is always online.